

Adaptive Image Enhancement for Robust Object Detection Under Adverse Weather Conditions

1. Introduction / Overview

Modern object detectors such as YOLO perform well in normal driving environments but can lose accuracy under rain, fog, night-time and low-visibility conditions. These conditions reduce contrast, introduce noise, blur object boundaries and make objects harder to detect. This research investigates whether adaptive image enhancement before YOLO detection can improve object detection performance.

Core research question: Can adaptive image enhancement improve YOLO-based object detection performance under adverse environmental conditions?

2. Motivation

- Adverse weather and illumination conditions can degrade camera images.
- Rain can create streaks, reflections and blurred boundaries.
- Fog reduces contrast and visibility, especially for distant objects.
- Night scenes commonly suffer from low illumination, noise and poor contrast.
- A visually enhanced image does not necessarily produce better object detection.
- The research therefore focuses on downstream detection performance rather than image appearance alone.

3. Background Study / Literature Review

Paper / Study	Main Idea	Relevance
Image-Adaptive YOLO (Liu et al., 2022)	Jointly optimizes image processing and YOLO detection for adverse conditions.	Provides a strong basis for adaptive enhancement.
Fog-Aware Adaptive YOLO	Uses an adaptive strategy according to foggy or normal conditions.	Supports condition-aware image processing.
Multi-module Enhanced Low-Light Object Detection (2026)	Combines low-light enhancement with YOLO-based detection.	Relevant to night-time object detection.
YOLO-D: Domain Adaptive Low-Light Detection	Combines low-light enhancement and domain adaptation with YOLOv8.	Shows low-light detection as a domain-shift problem.
Detection-Oriented Deraining (Chang & Yang, 2025)	Optimizes deraining for object detection instead of visual quality alone.	Supports detection-oriented enhancement.
DCP-Based Dehazing + Lightweight U-Net (2026)	Uses dehazing as preprocessing and measures its effect on object detection.	Relevant to fog and dehazing experiments.
BDD100K (Yu et al., 2020)	Large-scale driving dataset containing object, weather, scene and time-of-day information.	Provides the primary experimental dataset.

4. Gap Analysis

- Many studies focus on one degradation, such as rain, fog or low light, rather than a unified condition-aware framework.
- Image quality improvement does not always translate into better object detection.
- A single enhancement method may not be optimal for all environmental conditions.
- BDD100K enables condition-wise evaluation using weather and time-of-day metadata.
- BDD100K does not have an explicit 'poor visibility' label; a derived subset should be created using metadata and measurable image-quality or visibility criteria.

5. Novelty / Contribution

- **Condition-aware enhancement:** Select an enhancement strategy according to environmental conditions.
- **Multiple adverse conditions:** Evaluate rain, fog, night and derived low-visibility subsets within one framework.

- **Condition-wise evaluation:** Report detection performance separately for each condition.
- **Detection-oriented evaluation:** Judge enhancement primarily by its effect on YOLO detection performance.

6. Dataset

Dataset: BDD100K

Hugging Face source: [dgural/bdd100k](https://huggingface.co/datasets/dgural/bdd100k)

Link: <https://huggingface.co/datasets/dgural/bdd100k>

Relevant metadata: Weather, time-of-day, scene categories and object-detection annotations.

Relevant object classes: car, bus, truck, person, rider, bike, motor, traffic light, traffic sign and train.

7. Proposed Methodology

- Prepare BDD100K images and detection annotations.
- Group images using weather and time-of-day metadata.
- Create normal and adverse-condition subsets.
- Apply condition-specific enhancement methods: deraining for rain, dehazing for fog and low-light enhancement for night scenes.
- Run YOLO on original images as the baseline.
- Run YOLO on enhanced images.
- Run YOLO with the proposed adaptive enhancement pipeline.
- Compare performance overall and separately for each condition.

8. Experimental Design

Experiment	Pipeline
Baseline	Original BDD100K → YOLO
Fixed Enhancement	BDD100K → One Enhancement Method → YOLO
Adaptive Enhancement	BDD100K → Condition Identification → Appropriate Enhancement → YOLO

9. Evaluation Metrics

- mAP@50
- mAP@50:95
- Precision
- Recall
- F1-score
- Inference time / FPS (optional)
- Computational cost (optional)

10. Condition-wise Evaluation

Condition	Baseline YOLO	Enhanced YOLO	Adaptive YOLO
Clear / Normal	—	—	—
Rainy	—	—	—
Foggy	—	—	—
Night	—	—	—
Dawn/Dusk	—	—	—
Overall	—	—	—

11. Research Hypotheses

- H1: Adaptive image enhancement improves YOLO detection performance under adverse conditions.
- H2: Condition-specific enhancement performs better than applying one enhancement method to all images.
- H3: Enhancement provides greater benefit under adverse conditions than under normal conditions.
- H4: Detection-oriented enhancement can improve detection without necessarily maximizing conventional image-quality metrics.

12. Conclusion

This research investigates whether adaptive image enhancement can make YOLO object detection more robust under adverse driving conditions. BDD100K will be used to compare original, enhanced and adaptively enhanced images across rain, fog, night and other relevant conditions. The primary goal is to determine when enhancement is useful, which enhancement strategy is appropriate, and how much it improves downstream object detection.